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Power Supply:

GEL BATTERIES Are Much SMALLER In Terms Of SIZE And FOOTPRINT

We have seen an increase in the number of electric rickshaws on the streets recently. One company has identified a problem that all such vehicles are facing in their power supply. Biju Bruno, managing director, Greenvision Technologies, speaks with Dilin Anand from EFY about it

Q. What did your market research reveal about the current state of batteries in rickshaws, and what do you do about it?

A. After taking feedback from rickshaw owners, we realised that traditional batteries end up getting replaced every six months. That is a lot of waste. A place like New Delhi alone has 250,000 rickshaws including unregulated ones.

We developed gel based, maintenance-free batteries for electric vehicles, which have been especially designed for Indian conditions. In electric rickshaws, these batteries can provide a range of 80 kilometres with overnight charging and offer a longer battery life cycle.

Q. Any specific challenges in electric rickshaws that made it difficult for traditional batteries to perform?

A. The charger provided with the e-rickshaw is poorly designed in most cases and leads to over-charging of the battery and consequent water loss. This requires water topping up at frequent intervals, sometimes even weekly. The batteries need to be topped up with demineralised water but rickshaw drivers typically use tap water, which leads to shorter battery life.

The second challenge is the long duty cycle of 80km to 100kms daily in poor road conditions. In every e-rickshaw there are four heavy batteries that are not properly anchored. This leads to failures in inter-cell welds.

Then, the state of charge drops as the day progresses. If the battery is not able to maintain the minimum required charge, it is not functional.

The third is that the high temperatures we see now hovering at around 35 degrees (even 45 degrees at times) increases the water loss even more. When this happens, the plate gets corroded and that initiates a vicious cycle of deteriorating batteries. These are sorted out by gel based batteries.

Q. Any parallels between the challenges here and other sectors?

A. Higher temperatures and a partial state of charge means that solar and electric vehicles have similar challenges and therefore advantages you design into one battery solution can be easily ported to battery solutions for the other. The electrolyte and the separator are two elements that have been innovated for the gel battery.

Q. Could you elaborate more about gel based battery technology?

A. Electrolyte is a gel, and battery acid is mixed with pyrogenic silica. Gel by nature is hygroscopic, so it can absorb water from anywhere. This means that there is no water loss, which usually happens when water molecules break down in high temperatures.

The separator is what is usually placed in between the positive and negative plates. We use specially-

designed separators that have a very long life. In our testing, the batteries continued to work for three years at elevated temperatures.

Q. Are there any manufacturing challenges for gel based batteries?

A. Production process in terms of battery plates is similar to that of maintenance-free batteries, but the formula used for the paste is very different. The alloy used in production is also a slightly different version of what you usually see. Overall, we use the same equipment but the parameters change. When it comes to the assembly side, I cannot explain in detail right now because we have applied for a patent.

Q. Any other sectors apart from automotive and solar where gel based batteries could leave a mark?

A. One of the biggest segments that I see is home based inverters. For most inverters, battery used is tubular. Gel based batteries come out to be much smaller and maintenance-free—the size would be approximately half the size of a regular lead-acid battery, so you can double the battery capacity in the same space.

We have also developed prototypes for home lighting systems, where we have designed gel based batteries as replacements with a smaller footprint.

We are also working on a lithium-ion power pack for electric cars. **EFY**